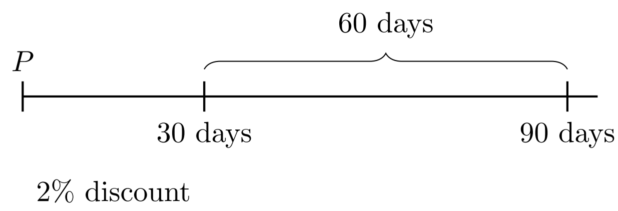
Problem 5.16

The terms of payment of a certain debt are: net cash in 90 days or 2% discount for cash in 30 days. What rate of simple interest is earned if the discount is taken advantage of?



Let the amount of the debt be P.

A 2% discount means the amount saved is

$$0.02P.$$

If the discount is taken, the amount actually paid at 30 days is

$$P - 0.02P = 0.98P.$$

Taking the discount means saving $0.02P$ by paying $0.98P$ 60 days early, since

$$90 - 30 = 60 \text{ days}.$$

We use the simple interest formula:

$$I = Prt.$$

Here,

$$0.02P = (0.98P) \cdot r \cdot \frac{60}{360}.$$

Solving for r:

$$\frac{360(0.02P)}{60} = 0.98P\,r.$$

So,

$$0.12P = 0.98P\,r.$$

Therefore,

$$r = \frac{0.12}{0.98} \approx 0.122448.$$

Thus, the simple interest rate earned is

$$\boxed{12.2448\%}.$$